Electricity Supply Shocks and Manufacturing: Evidence from the Fukushima Nuclear Accident

Daiji Kawaguchi
U Tokyo

Takanao Tanaka
UC Berkeley

UC Berkeley
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Electricity Supply Shocks

- Electricity is becoming increasingly important in developed economies
  - Demand growth: electrification and data centers
- At the same time, electricity supply remains vulnerable to shocks
  - Geopolitical risks, clean energy transitions
  - Supply disruptions can manifest in various ways
- We know little about large-scale, persistent electricity supply shocks
  - Most evidence comes from oil shocks or small, temporary electricity price changes
- How do large-scale, persistent electricity supply shocks affect industry?

How do large-scale, persistent electricity supply shocks affect industry?

- **Idea:** Electricity supply shock following Fukushima and the nuclear phase-out
  - Large-scale, nationwide, persistent
  - Raised electricity prices and led to an electricity demand reduction policy
- **Data:** Plant-level Census of Manufactures, 2005–2018
- **Approach:**
  1. *Reduced-form*: Regional variation in shocks
     → High nuclear reliance and low initial reserve capacity → severe shocks
  2. *Production function*: Electricity as an input
     → Replicate the reduced-form impacts: mechanism and counterfactual

Preview of Findings

- **Lage Impacts:** Electricity use falls 6.7%; output falls 1.7%
  - $\approx$ 4.9 trillion yen or USD 50 billion annually
- **Limited plant responses:**
  - Product mix, outsourcing, fuel use, within-firm reallocation, or exit
- **Puzzle:** Production function understates the impacts (Cobb-Douglas: 0.4% output loss)
  - Electricity expenditures: only 2% of plant revenue
- **Mechanisms:** Input complementarity and capital idleness
- **Counterfactuals:** Higher electricity prices explain most output losses

What Does This Paper Add?

1. **Electricity and industry in developed countries** (Kahn and Mansur, 2013; Bijnens et al., 2022; Wolverton et al., 2022; von Graevenitz and Rottner, 2024)
   ⇒ We study large-scale, persistent shocks in electricity markets

2. **Oil shocks** (Kilian, 2008; Hamilton, 2013; Bernanke et al., 1997; Baqaee and Farhi, 2019)
   ⇒ Microdata to understand plant responses
   ⇒ Quasi-experimental variations: identify substitution parameters and capital idleness

3. **Nuclear policies** (Davis and Hausman, 2016; Severnini, 2017; Neidell et al., 2021; Jarvis et al., 2022)
   ⇒ We investigate downstream effects on industrial production

Fukushima Nuclear Accident and Electricity Supply Shock

Great East Japan Earthquake and Fukushima Accident

- On March 11, 2011; a magnitude-9.0; 4th largest since 1900; 20,000 deaths
- Tsunami disabled cooling at Fukushima Daiichi, causing three reactor meltdowns





Earthquake and tsunami



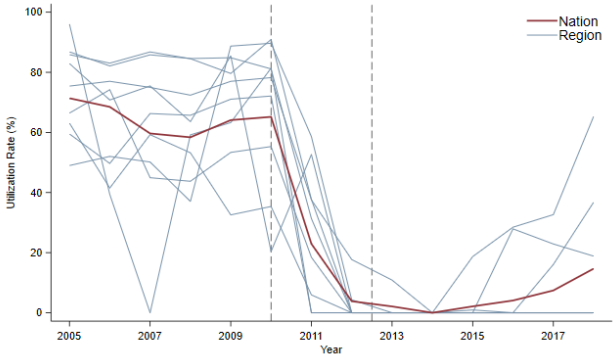


Fukushima Daiichi nuclear accident

Nuclear Shutdown

- Concerns about nuclear safety led to a persistent nuclear phase-out
- Nuclear accounted for 30% of supply

Utilization Rate of Nuclear Power Plants



Electricity Supply Shock

Summer Peak-Day Demand and Planned Supply (GW), 2011

No.	Region	Demand 2010	Planned Supply 2011	Transmission (% of supply)	(Supply - Demand) / Demand 2010
1	Hokkaido	5.1	5.5	-0.6 (-10.9%)	8.5%
2	Tohoku	15.6	13.8	1.6 (11.6%)	-11.5%
3	Tokyo	59.9	54.7	-0.4 (-0.7%)	-8.7%
4	Hokuriku	5.7	5.8	0.0 (0.0%)	1.8%
5	Chubu	27.1	28.0	0.0 (0.0%)	3.3%
6	Kansai	31.0	29.4	0.8 (2.7%)	-5.2%
7	Chugoku	12.0	12.4	0.7 (5.6%)	3.3%
8	Shikoku	6.0	6.2	0.0 (0.0%)	3.3%
9	Kyushu	17.5	17.9	0.0 (0.0%)	2.3%

- Supply $<$ pre-accident demand Reserve margin $\leq 5\%$—“critical”
- Typically, transmission $<$ a few %

Electricity Supply Shock

Summer Peak-Day Demand and Planned Supply (GW), 2012

No.	Region	Demand 2010	Planned Supply 2012	Transmission (% of supply)	(Supply - Demand) / Demand 2010
1	Hokkaido	5.1	4.9	0.0 (0.0%)	-4.2%
2	Tohoku	15.6	14.8	-0.1 (-0.7%)	-5.3%
3	Tokyo	59.9	57.7	-0.3 (-0.5%)	-3.8%
4	Hokuriku	5.7	5.8	-0.1 (-1.7%)	0.8%
5	Chubu	27.1	27.9	-1.1 (-3.9%)	2.8%
6	Kansai	31.0	27.8	1.2 (4.3%)	-10.2%
7	Chugoku	12.0	12.4	-0.5 (-4.0%)	2.8%
8	Shikoku	6.0	5.9	-0.2 (-3.4%)	-1.6%
9	Kyushu	17.5	15.7	0.5 (3.2%)	-10.0%

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Electricity Supply Shock

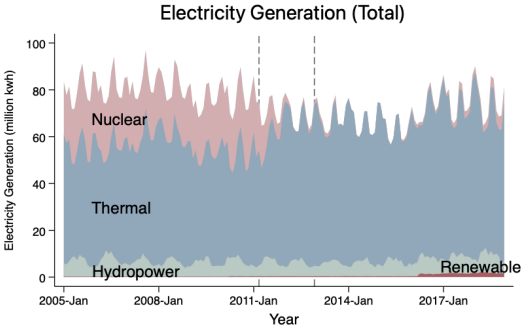
Summer Peak-Day Demand and Planned Supply (GW), 2013–2018 Average

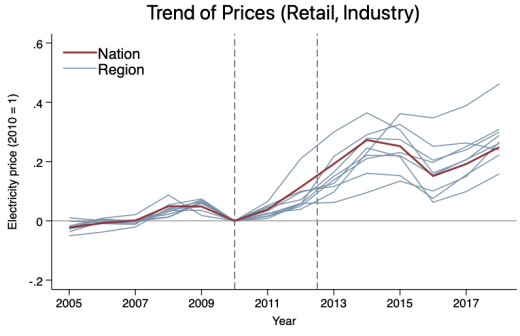
No.	Region	Demand 2010	Average Planned Supply 2013–2018	Transmission (% of supply)	(Supply - Demand) / Demand 2010
1	Hokkaido	5.1	5.2	-0.1 (-1.1%)	2.0%
2	Tohoku	15.6	15.1	-1.1 (-7.1%)	-2.9%
3	Tokyo	59.9	56.4	0.2 (0.3%)	-5.8%
4	Hokuriku	5.7	5.8	-0.1 (-1.6%)	1.8%
5	Chubu	27.1	27.6	-1.0 (-3.8%)	1.7%
6	Kansai	31.0	28.9	0.7 (2.6%)	-6.7%
7	Chugoku	12.0	12.4	-0.4 (-3.5%)	3.0%
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- Typically, transmission $<$ a few %

Nuclear Shutdown, Rationing, and Increased Prices

- Demand-side: Rationing, including an information campaign
- Supply-side: Use of retired/new thermal generation → increased prices





Data

Plant Outcomes Linked to Regional Electricity Supply Shocks

1. Census of Manufactures (CMF), 2005–2018
   - Plant-level administrative data; inputs and outputs
   - Covers 80% of total manufacturing output
   - Product mix, firm identifiers, entry and exit
2. Energy Consumption Statistics (ECS), 2007–2018
   - Plant-level administrative data; physical quantities of fuel inputs
   - Fuel used for production and on-site electricity generation
3. Regional electricity supply–demand data, 2005–2018
   - Peak-day demand and planned peak-day supply

Summary statistics

1. How do large electricity supply shocks affect industry?

Region-year supply shocks ≈ required demand cuts

We use peak-day demand/supply to construct region-year ($r - t$) supply shocks.

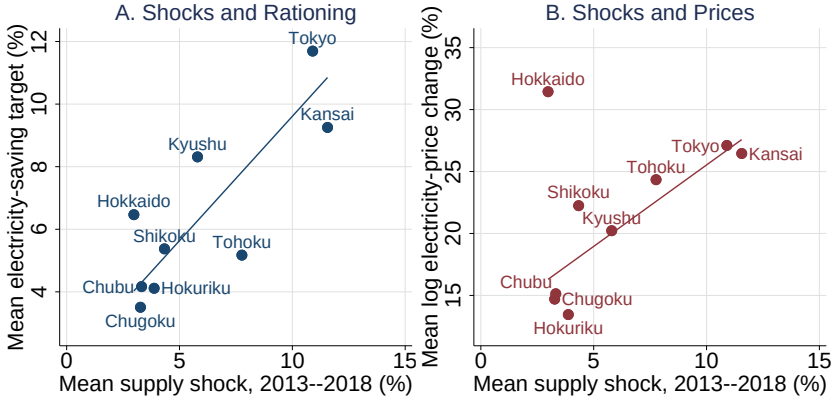
$$\mathcal{S}_{rt} = \Bigg(\underbrace{1.05\, D^{\text{peak}}_{r,2010}}_{\substack{\text{required} \\ \text{supply}}} - \underbrace{S^{\text{plan,peak}}_{rt}}_{\substack{\text{available} \\ \text{supply}}} \Bigg) \underbrace{\frac{1}{D^{\text{peak}}_{r,2010}}}_{\substack{\text{normalized by} \\ \text{regional demand}}} \quad , \quad \text{if available} < \text{required supply.} \qquad (1)$$

- Required supply: pre-accident peak demand ($D^{\text{peak}}_{r,2010}$) plus a 5% reserve margin
- Available supply: planned peak-day supply

Shock trends and maps

Supply Shocks Track Saving Targets and Prices

- Summer saving target for demand curtailment policies (He and Tanaka, 2023)
- Isolate price changes due to shocks from global trend



Regression table

Compare regions with large shocks and low-shocks

Estimation

$$\ln Y_{ijrt} - \ln Y_{ijr2010} = \underbrace{\rho_1 \mathcal{S}_{rt} \mathbb{1}\{2011 \le t \le 2012\}}_{\text{phase-out}} + \underbrace{\rho_2 \mathcal{S}_{rt} \mathbb{1}\{2013 \le t \le 2018\}}_{\text{post-phase-out}} + \mu_{jt} + \epsilon_{ijrt}. \quad (2)$$

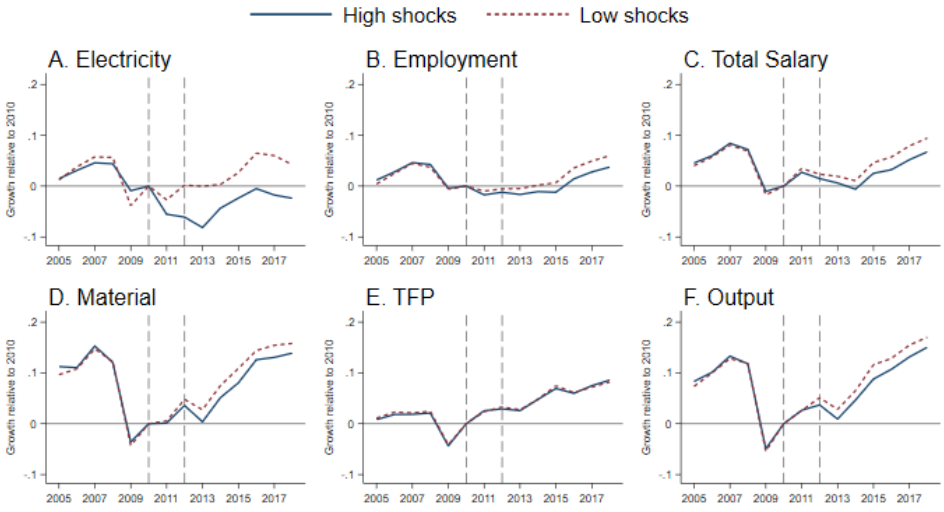
- i: plant; j: industry; r: region; t: year
- Y: output, electricity, labor, materials, or TFP; μ_{jt}: industry–year fixed effects

Instrumental variables

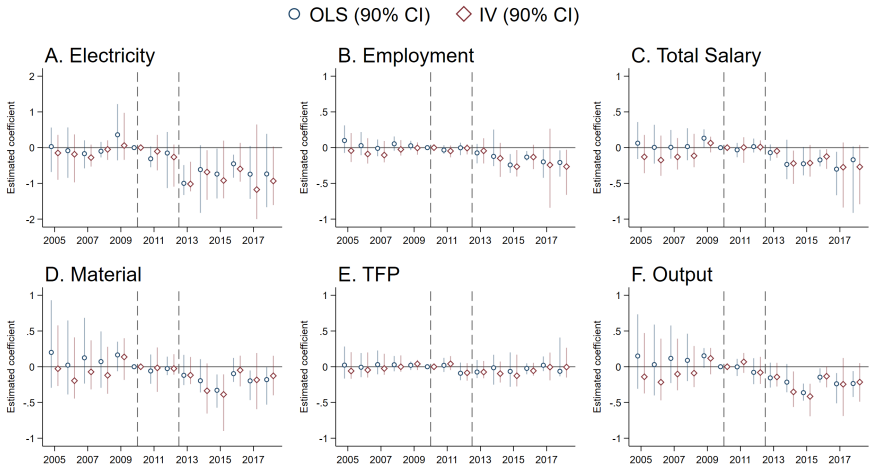
$$\mathbf{Z}_{rt} = \left(Sh^{\text{nuclear}}_{r,2010} \times g^{\text{nuclear}}_{t,-r},\ Sh^{RM}_{r,2010} \right). \quad (3)$$

- High nuclear reliance + low initial reserve margin → More severe shock First stage
- $Sh^{\text{nuclear}}_{r,2010}$: 2010 nuclear share; $g^{\text{nuclear}}_{t,-r}$: national growth rate; $Sh^{RM}_{r,2010}$: 2010 reserve margin

Outcome Trends before and after the Fukushima Accident

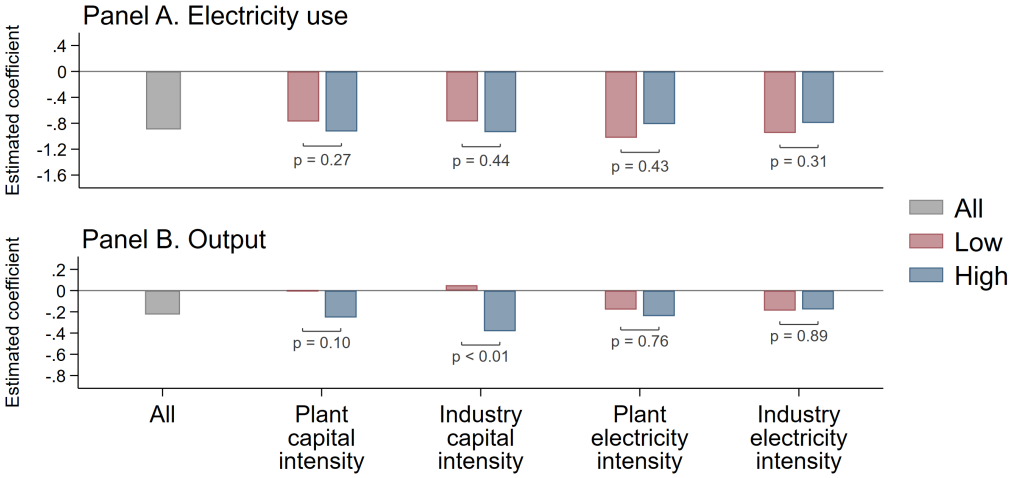


Electricity Supply Shocks Reduce Inputs and Output



- Nine regions: 90% CI wild cluster bootstrap clustered by region.

Larger Output losses among Capital-intensive Plants/Industries

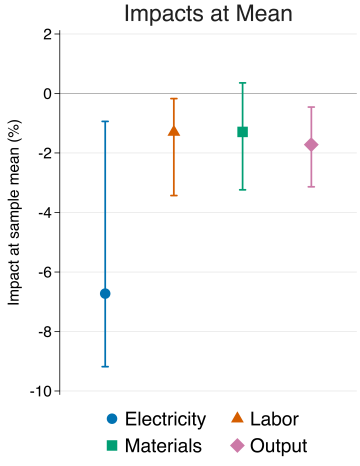


Robustness Checks

- **Alternative explanations** Results
  - Supply-chain disruptions: direct and indirect exposure
  - Local demand: regional activity and population

- **Alternative specifications** Results
  - Predetermined controls: 2010 plant size and composition
  - Fixed effects: plant and year fixed effects
  - Weighting: equal plant weights

How do large-scale electricity supply shocks affect industry?



- Shocks reduce electricity by 6.7% and output 1.7%
  - 4.9 trillion JPY or USD 50 billion annually
  - $\approx$ 3 years of manufacturing growth ($0.5\% \times 3$)
- Modest electricity price changes often cause no output contractions (e.g., Wolverton et al., 2022; von Graevenitz and Rottner, 2024)
- Comparable to past energy shocks ($\approx$ 0.5–3% of GDP) (e.g., Kilian and Vigfusson, 2017)
- Outages cause much larger impacts (e.g., Allcott et al., 2016; Cole et al., 2018)

Robustness

2. How do plants respond to electricity supply shocks?

Do Plants Switch Products or Increase Outsourcing?

	Product-level		Outsourcing	
	Switching (1)	Electricity Intensity (2)	Out-sourcing (3)	Share in materials (4)
IV Estimates				
Supply Shocks	0.07	0.02	-0.02	-0.05
(Phase-out)	[0.33]	[0.32]	[0.95]	[0.51]
Supply Shocks	0.07	0.03	-0.36	-0.08
(Post-phase-out)	[0.52]	[0.44]	[0.37]	[0.46]
Impacts at Mean	0.5%	0.2%	-2.7%	-0.6%

- Plants may reduce electricity intensity for each product, or buy intermediate goods, rather than producing them.
- Little evidence of product switching or increased outsourcing

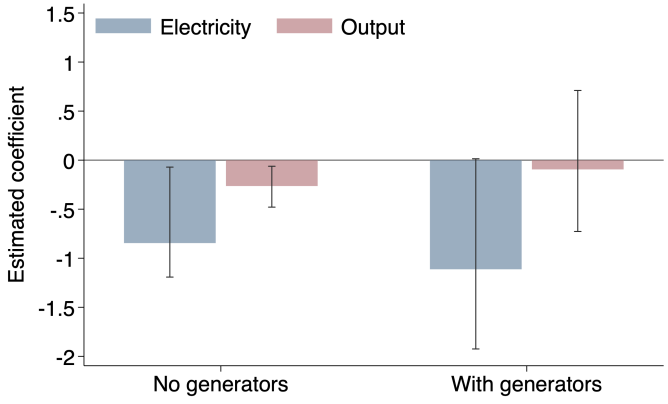
Is Fuel a Substitute for Electricity?

Dependent Variable: $\Delta \ln(Y) = \ln(Y) - \ln(Y_{2010})$	Fuel consumption		Self-generation	
	For production (1)	For self-generation (2)	Intensive margin (3)	Extensive margin (4)
IV Estimates				
Supply shocks (Post-phase-out)	-0.12 [0.35]	1.33 [0.19]	6.46*** [0.00]	0.23*** [0.01]
Supply shocks (Phase-out)	-0.35 [0.53]	3.88** [0.04]	4.92* [0.10]	0.30*** [0.00]
Impacts at Mean	-2.6%	29%	36%	2.2%

- (1) Shocks do not increase fuel used in production
- (2) & (3) Shocks increase self-generation among plants initially owning generators
- (4) Extensive-margin impacts are 2.2%

Plants initially with Generators Experience No Output Losses

- Generator ownership is 8% at baseline
- Generator-owned plants reduce electricity purchased from grid



Working Hours: Modest Medium-Run Declines

Dep Var (hours per worker): $\Delta \ln(H)$ relative to 2010	Total hours (1)	Scheduled hours (2)	Overtime (3)
Effects of Supply Shocks using 2SLS			
Supply Shocks	-0.12*	-0.10**	-0.17
(2013–2018)	[0.094]	[0.029]	[0.713]
Year FE	Yes	Yes	Yes
Observations	552	552	552

Source: MHLW, Monthly Labour Survey (manufacturing establishments with 30+ workers).

How do plants respond to electricity supply shocks?

Over 3–7 years, we find NO evidence that plants

- Switch products or increase outsourcing
- Increase fuel use directly in production
- Reallocate production within the same firm link
- Relocate or exit link

Self-generation helps avoid output losses, but less than 10% of plants use it

Shocks did not induce notable adjustments, potentially explaining large output losses

Summary of Reduced-Form Results

- Shocks reduce manufacturing inputs and output
  - Electricity use: 6.7%; output: 1.7%; labor and materials: 1.3% each
  - Larger output losses at capital-intensive plants
- No significant effects on product switching, outsourcing, or production fuel use
- Shocks modestly reduce working hours per worker (medium run)
- We develop a model to
  - Explain mechanisms: how substitutable is electricity?
    Why larger losses at capital-intensive plants?
  - Quantify channels: pricing versus moral suasion?
  - Evaluate policies: what could mitigate output losses?

3. What mechanism can replicate the observed output losses?

Step 1: Estimate CES production functions-Approach

$$Y_{it} = \Omega_{it}\left(\alpha_K K_{it}^{\rho} + \alpha_L L_{it}^{\rho} + \alpha_M M_{it}^{\rho} + \alpha_E E_{it}^{\rho}\right)^{\frac{1}{\rho}}, \qquad \sigma \equiv \frac{1}{1-\rho}. \tag{4}$$

Estimate output elasticities θ_X following Gandhi, Navarro, and Rivers (2020) link

- Flexible inputs (E, M): identify θ_E and θ_M using expenditure shares
- Quasi-fixed inputs (L, K): Identify θ_L and θ_K using a control function

Calibrate the CES weights $\alpha_K, \alpha_L, \alpha_M, \alpha_E$ given σ:

- Normalize the estimated $\theta_K, \theta_L, \theta_M, \theta_E$ to obtain baseline input shares s_K, s_L, s_M, s_E
- Choose $\alpha_K, \alpha_L, \alpha_M, \alpha_E$ so that the CES model reproduces these baseline shares

Estimation by industry (153) using 2005-2010 data

Step 2: Predict outcomes under observed and baseline scenarios

Two shock channels

- "Rationing" on peak-season (summer) electricity use: $E^S_{i,post} \leq (1 - R)E^S_{i,base}$
- Shock raises electricity prices from $\hat{p}^E_{base} = 1$ to $\hat{p}^E_{post}$ regression
- Model-implied output impact: $Y(\hat{p}^E_{post}, R) - Y(1, 0) \rightarrow E, M, K, L$

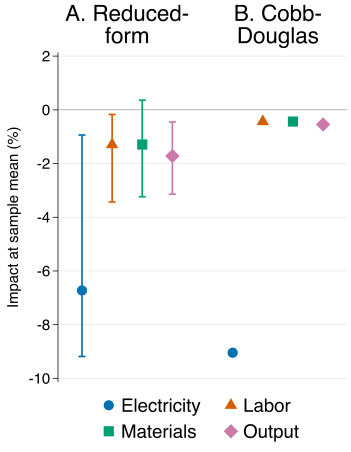
Assume isoelastic demand: $Y_i = A_i P_i^{-\eta}, \quad \eta = 5$

Predictions:

- Same sample; 2010 baseline, 2013–18 post period; all inputs adjustable.
- Hold production technology, other input prices, and demand elasticity fixed
- Predict peak and off-peak seasons separately, then aggregate to annual outcomes
- Plants with generators; shocks reduce electricity use but do not affect other inputs and output

Prediction

Two Discrepancies between Reduced-form Results and Cobb–Douglas Model



Cobb–Douglas prediction ($\sigma = 1$)

Discrepancy 1: Relative input declines differ

- Electricity: 6.7% vs 8.9% (CD)
- CD allows flexible substitution
- Substitution limits declines in other inputs (Material, Labor: 1.3% vs 0.4% (CD))

Discrepancy 2: Smaller output losses

- 1.7% (reduced-form) vs 0.4% (CD) in output
- Echoes the “oil puzzle” in the oil-shock literature (Kilian, 2008; Hamilton, 2013)

Two Mechanisms: Capital Idleness and Complementarity

1. Allow σ to vary: stronger complementarity among electricity, materials, and labor

2. Electricity-induced capital idleness

   Idea: furnaces cannot operate without electricity
   - Electricity unavailability can lower capital service flow ($\tilde{K}_{it}$), given installed capital (K_{it})
   - $\neq$ substitution, which alters K_{it}

   Motivation
   - Well established in oil shock papers (Finn, 2000; Atkeson and Kehoe, 1999)
   - Larger output losses among capital-intensive plants and industries
   - Qualitative documents supporting equipment stoppage
   - Working hours decline modestly in the medium run Working hours

Cost-share intuition

Two Mechanisms: Capital Idleness and Complementarity

Model: Capital services $\widetilde{K}_{it}$, rather than installed capital K_{it}, enter production

$$Y_{it} = \Omega_{it}\left(\alpha_K \widetilde{K}_{it}^{\rho} + \alpha_L L_{it}^{\rho} + \alpha_M M_{it}^{\rho} + \alpha_E E_{it}^{\rho}\right)^{1/\rho},$$

$$\widetilde{K}_{it} = \left[(1-\kappa)K_{it}^{\rho} + \underbrace{\kappa}_{\substack{\text{share of capital} \\ \text{requiring electricity}}} (v_i E_{it})^{\rho}\right]^{1/\rho}, \qquad v_i = \frac{K_{i,2010}}{E_{i,2010}}.$$

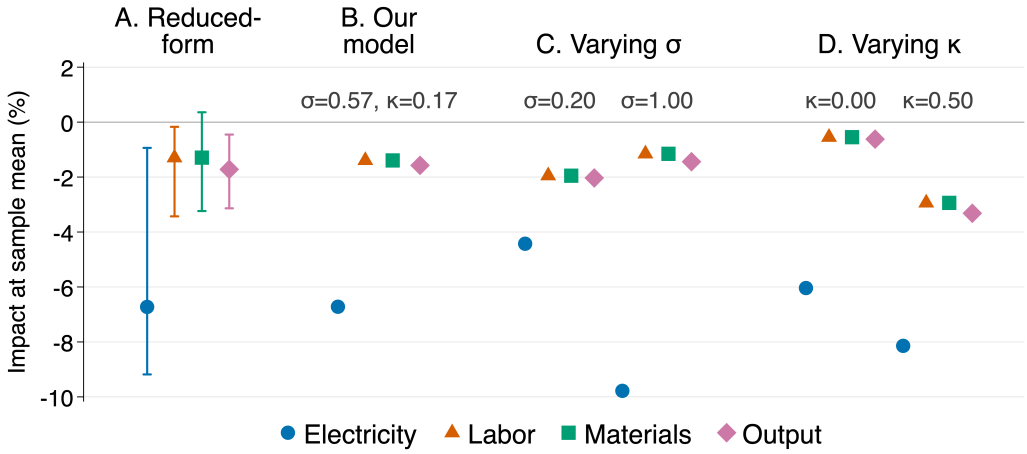
Interpretation:

- A 1% reduction in electricity idles κ% of installed capital at baseline
- Gives more weight to E and less to K.

Estimation: Jointly choose $\hat{\kappa}$ and $\hat{\sigma}$ to match reduced-form impacts on E, L, M, Y

Cost-share intuition

Two Mechanisms: Complementarity ($\sigma = 0.57$) and Capital Idleness ($\kappa = 0.17$)



What mechanism can replicate the observed output losses?

- Cobb-Douglas cannot replicate reduced-form impacts ≈ Oil Puzzle
  - Electricity expenditures: only 2% of plant revenue
- We incorporate elasticity of substitution and capital idleness:
  - $\widehat{\sigma} = 0.57$, within the 0.2–0.8 range in prior estimates (van der Werf, 2008; Bretschger and Jo, 2024; Hawkins-Pierot and Wagner, 2026)
  - $\widehat{\kappa} = 0.17 \Rightarrow$ 6.7% reduction in electricity idles 1.14% of installed capital
  - We use micro-data and quasi-experimental variation for estimation, in contrast to the macro literature (Finn, 2000)
- Suggesting asymmetry between electricity price increases and decreases

4. What policies could mitigate output losses?

Pricing or Moral Suasion: Which Matters More?

- Moral suasion or pricing: which accounts for more of the manufacturing output loss?
  - Little evidence of product switching, outsourcing, or fuel substitution
  - Identifying the dominant channel guides policies to mitigate losses
- If moral suasion matters more: who should be targeted?
  - Users who can conserve electricity with smaller output losses
- If prices matter more: what drives Japan's electricity prices?
  - Regulation and regional segmentation can raise supply costs
  - Evaluate regional market integration and reserve sharing

Pricing accounts for 70-93% of input/output losses

$$\underbrace{Z(\hat{p}^E_{post}, R) - Z(1,0)}_{\text{Total impact}} = \underbrace{Z(1,R) - Z(1,0)}_{\text{Rationing first}} + \underbrace{Z(\hat{p}^E_{post}, R) - Z(1,R)}_{\text{Pricing second}}$$

$$\text{or} \qquad = \underbrace{Z(\hat{p}^E_{post}, 0) - Z(1,0)}_{\text{Pricing first}} + \underbrace{Z(\hat{p}^E_{post}, R) - Z(\hat{p}^E_{post}, 0)}_{\text{Rationing second}}.$$

	Electricity		Output	
	Level (%)	% of total	Level (%)	% of total
Total impact	−6.7		−1.6	
= Rationing first	−2.0	29.7	−0.5	28.8
+ Pricing second	−4.7	70.3	−1.1	71.2
or				
= Pricing first	−6.2	92.7	−1.5	93.0
+ Rationing second	−0.5	7.3	−0.1	7.0

Own price elasticity ≈ 0.8, Output elasticity ≈ 0.2

Outcome	Reduced-form	Structural model
Electricity use (own-price)	−0.80	−0.74
Labor use (cross-price)	−0.15	−0.15
Materials use (cross-price)	−0.15	−0.15
Output	−0.20	−0.17

- Structural estimates only use price → smaller
- Own price elasticity in the range of the prior literature (−0.09 to −1.5) (Gerster and Lamp, 2024; von Graevenitz and Rottner, 2024; Martin et al., 2014)

Elasticity calculation

Market integration recovers nearly one-third of output losses

- Integration lets lower-cost generators serve demand across regions and enables reserve sharing
- National mean log-price increase falls from 8.4% to 5.7% (−2.8 pp)

	Electricity	Output	Annual output loss (JPY billion)
Reduced-form impact	−6.6%	−1.7%	4,920
Price-only baseline	−6.2%	−1.5%	4,231
Market integration	−4.2%	−1.0%	2,864

What policies could mitigate output losses?

- Higher electricity prices account for most input and output losses
  - Pricing explains 70–93% of the losses; rationing explains 7–30%
- Plants respond substantially to persistent electricity price increases
  - A 10% price increase reduces electricity use by $\approx 8\%$ and output by $\approx 2\%$
- Regional market integration mitigates these losses
  - The mean price increase falls from 8.4% to 5.7%
  - Price-driven output losses fall by 32% (JPY 4.2 to 2.9 trillion annually)

5. Conclusion

Electricity-market shocks create large industrial losses

- **Impacts:** Significant output losses
  - Electricity use falls 6.7%; output falls 1.7% ($\approx$ JPY 4.9 trillion annually)
- **Plant responses:** Little impact on product mix, outsourcing, fuel use, reallocation, or exit
  - Self-generation helps, but fewer than 10% of plants use it
- **Mechanisms:** Capital idleness and input complementarity
  - Cobb–Douglas understates the reduced-form impacts
  - A 1% electricity decline idles 0.17% of capital services; $\widehat{\sigma} = 0.58$
- **Counterfactuals:** Higher prices explain most output losses
  - Regional market integration reduces output losses by 32%

Thank you! Takanao Tanaka takanao_tanaka@berkeley.edu | takanao-tanaka.com

Appendix

Summary Statistics

	Mean	S.D.
Panel A. Plant Characteristics		
Revenue (million yen)	4,999	11,313
Employment (headcount)	127	166
Capital stock (million yen)	1,035	2,485
Investment (million yen)	185	532
Export share (%)	2.8	11
Multi-plant firm (yes = 1)	0.42	0.49
Panel B. Revenue shares		
Labor	0.22	0.14
Capital	0.033	0.042
Materials	0.51	0.20
Electricity	0.020	0.023
Panel C. Plant Energy Use		
Fuel input for production (TJ)	30	114
Fuel input for self-generation (TJ)	4.2	31
Self-generated electricity (TJ)	3.6	24
Grid electricity purchased (TJ)	20	55
Self-generation ownership (yes = 1)	0.13	0.34

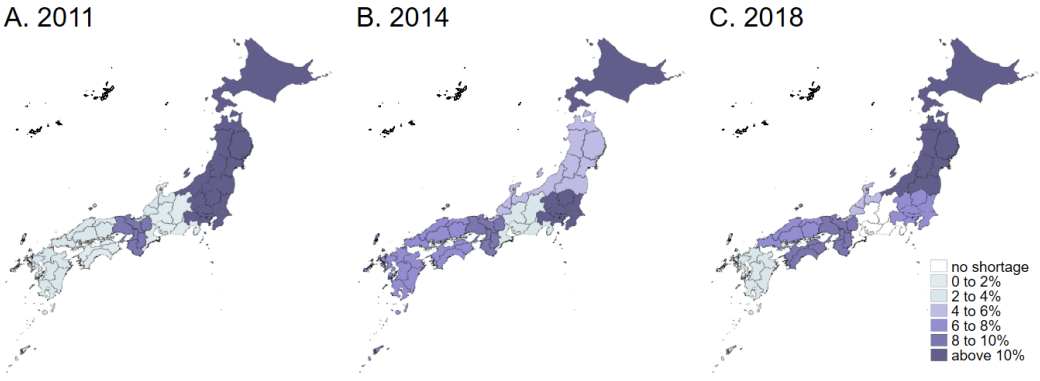
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Regional Electricity Supply Shocks over Time



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Electricity Supply Shocks across Regions and Years



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Supply Shocks Raised Saving Targets and Prices

$$Z_{rt} = \beta \mathcal{S}_{rt} + \gamma_{jt} + \varepsilon_{ijrt}, \qquad Z_{rt} \in \{\text{saving target}, \Delta \ln p^{E}_{rt}\} \tag{5}$$

	Electricity price		Saving target (rationing)	
	2011–2012	2013–2018	2011–2012	2013–2018
OLS	0.03	0.78***	0.72***	0.52***
	(0.19)	(0.20)	(0.09)	(0.10)
2SLS	0.00	1.12**	0.65***	0.70***
	(0.32)	(0.35)	(0.10)	(0.15)
Industry-by-year FE	Yes	Yes	Yes	Yes

- Per 1 pp shock, 2SLS saving-target response: 0.65 pp in 2011–2012; 0.70 pp after 2012
- At the mean shock, the predicted 2013–2018 price increase is 8.4%

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First-stage Results—Nuclear phase-out causes regional supply shocks

	Supply shock	
	Phase-out (1)	Post-phase-out (2)
$Sh^{\text{nuclear}}_{r,2010} \times g^{\text{nuclear}}_{t,-r}$	-0.45	-0.26
	(0.00)***	(0.00)***
$Sh^{RM}_{r,2010}$	-1.05	-0.82
	(0.00)***	(0.00)***
Year FE	yes	yes
Observations	18	54
First-stage F	30	26

Interpretation

- Average post-phase-out effect: $18\% \times 97\% \times 0.26 = 4.4$ ppt
- A reserve margin 5 ppt below the mean increases the shock by 4.1 ppt
- Strong first stages: $F = 30$ and $F = 26$

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Robustness: Supply Chains and Local Demand

	Baseline	Supply-chain controls	Local-demand controls
Post-phase-out 2SLS estimates			
Electricity use	−0.89*	−0.88**	−0.75**
	[0.06]	[0.01]	[0.02]
Output	−0.23**	−0.20*	−0.23
	[0.04]	[0.06]	[0.12]

- Supply-chain controls capture direct and indirect input–output exposure
- Local-demand controls capture regional activity, population, construction, consumption, and labor-market tightness

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Robustness: Specifications and Weighting

	Baseline	Plant controls	Two-way FE	Equal weights
Post-phase-out 2SLS estimates				
Electricity use	-0.89^{*}	-0.82^{**}	-0.79^{***}	-0.78^{**}
	[0.06]	[0.02]	[0.01]	[0.02]
Output	-0.23^{**}	-0.20	-0.17	-0.24^{***}
	[0.04]	[0.19]	[0.35]	[0.00]

- Predetermined controls use 2010 plant characteristics
- Two-way fixed effects use the plant panel; equal weights give every plant the same weight

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Do Firms Reallocate Production Across Regions?

Dep Var: $\Delta \ln(Y) = \ln(Y) - \ln(Y_{2010})$	Electricity (1)	Output (2)
Effects of Supply Shocks using 2SLS		
Supply Shocks	0.19	-0.03
(Phase-out Terms)	[0.44]	[0.74]
Supply Shocks	0.84**	-0.04
(Post-Phase-out)	[0.04]	[0.80]
Year-by-Industry FE	Yes	Yes

- Examine plants in other regions within the same firm
- Electricity use rises at those plants
- No evidence of revenue reallocation across regions

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Do Supply Shocks Affect Entry and Exit?

Dependent variable: $\Delta \ln(Y)$ relative to 2009	Number plants (1)	Entry (2)	Exit (3)
IV Estimates			
Supply shocks	-0.34	1.08	0.39
	[0.27]	[0.56]	[0.44]
Industry-year FE	yes	yes	yes

- No significant effect on the number of plants, entry, or exit

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Production Function Estimation

Assumptions

- Plants are price takers; static FOCs hold for electricity and materials
- Timing: choose $(k_{it}, l_{it}) \rightarrow$ observe $\omega_{it} \rightarrow$ choose $(e_{it}, m_{it}) \rightarrow$ realize ϵ_{it}
- Productivity is first-order Markov: $\omega_{it} = h(\omega_{i,t-1}) + \eta_{it}$, with $\mathbb{E}[\eta_{it} \mid k_{it}, l_{it}] = 0$

First stage: flexible inputs (E, M)

$$\ln s_{h,it} \equiv \frac{P_t^h X_{h,it}}{P_{it} Y_{it}} = \ln \theta_{hj} - \epsilon_{it}, \qquad , h \in \{E, M\}.$$

- We use $\mathbb{E}[\epsilon_{it} \mid k, l, m, e, \omega] = 0$ to identify $\widehat{\theta}_E$, $\widehat{\theta}_M$

Second stage: labor and capital (L, K)

$$\widehat{\mathcal{Y}}_{it} \equiv y_{it} - \widehat{\theta}_E e_{it} - \widehat{\theta}_M m_{it}, \quad \Rightarrow \quad \widehat{\omega}_{it}(\theta_L, \theta_K) \equiv \widehat{\mathcal{Y}}_{it} - \theta_L l_{it} - \theta_K k_{it} - \widehat{\epsilon}_{it} \tag{6}$$

- We use $\mathbb{E}[\eta_{it} l_{it}] = \mathbb{E}[\eta_{it} k_{it}] = 0$ to identify $\widehat{\theta}_L$, $\widehat{\theta}_K$

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Output Elasticities → CES Weights

Estimated output elasticities imply baseline shares:

$$\widehat{s}_{hj} = \frac{\widehat{\theta}_{hj}}{\sum_{q\in\{K,L,M,E\}} \widehat{\theta}_{qj}}.$$

For a candidate elasticity of substitution σ, use the CES technology:

$$Y_{it} = \Omega_{it}\left(\sum_{h\in\{K,L,M,E\}} \alpha_{hj} X_{h,it}^{\rho}\right)^{1/\rho}, \qquad \rho = \frac{\sigma-1}{\sigma}.$$

Choose CES weights to reproduce the baseline shares:

$$\widehat{\alpha}_{hj}(\sigma) = \frac{\widehat{s}_{hj}^{1/\sigma}}{\sum_{q\in\{K,L,M,E\}} \widehat{s}_{qj}^{1/\sigma}}, \qquad p_h^0 = 1.$$

Step 1: Estimate CES production functions-Output Elasticities

- Estimate production function separately by 153 industries
- Use 2005-2010, pre-shock data
- OLS typically overstates output elasticities of flexible inputs (correlated with productivity)

	Structural Estimates			OLS
	Mean	p25	p75	Mean
	(1)	(2)	(3)	(4)
Materials (θ_M)	0.446	0.386	0.506	0.492
Electricity (θ_E)	0.015	0.008	0.017	0.068
Labor (θ_L)	0.354	0.259	0.432	0.423
Capital (θ_K)	0.139	0.083	0.170	0.042

Production function

Prediction Details (1/3): Demand and Input Choices

The CES unit-cost function is

$$c_i(p_K, p_L, p_M, p_E) = \frac{1}{\Omega_i}\left(\alpha_K^{\sigma} p_K^{1-\sigma} + \alpha_L^{\sigma} p_L^{1-\sigma} + \alpha_M^{\sigma} p_M^{1-\sigma} + \alpha_E^{\sigma} p_E^{1-\sigma}\right)^{\frac{1}{1-\sigma}}.$$

Optimal output and input choices

$$P_i^* = \frac{\eta}{\eta - 1} c_{ij}(\mathbf{p}), \qquad Y_i^* = B_i (P_i^*)^{-\eta} \qquad X_{ih}^* = Y_i^* \frac{\partial c_{ij}(\mathbf{p})}{\partial p^h} \tag{7}$$

For Rationing, we use shadow price;

$$\lambda_i^R \equiv \Delta \ln \widetilde{p}_i^E = \frac{\Delta \ln E_i^{S,R}}{\varepsilon_{E,i}}, \qquad \Delta \ln E_i^{S,R} = \ln(1 - \overline{R}_r), \qquad \varepsilon_{E,i} \equiv \frac{\partial \ln E_i}{\partial \ln p^E} < 0.$$

Then use this shadow price and the closed-form solution 7

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Prediction Details (2/2): Simulation Inputs

Object	Level	Description or source	Baseline	Post-phase-out
Panel A. Technology				
σ	Common	Elasticity of substitution	$\widehat{\sigma}$	$\widehat{\sigma}$
$\widehat{\boldsymbol{\theta}_j}$	Industry	Estimated output elasticities	$\widehat{\boldsymbol{\theta}_j}$	$\widehat{\boldsymbol{\theta}_j}$
Ω_i	Plant	Baseline productivity	Ω_i	Ω_i
G_i	Plant	Generator ownership	G_i	G_i
Panel B. Demand				
η	Common	Demand elasticity	5	5
B_i	Plant	Demand shifter; $Y_{i,2010}$	B_i	B_i
Panel C. Policies				
R_r	Region	Electricity-saving target	0	$\overline{R}_r$
ϕ	Common	Shock-season share of year	–	90/365
p_r^E	Region	Regional electricity price	1	$\overline{p}_r^E$
p^K, p^L, p^M	Common	Non-electricity input prices	1	1
Panel D. Capital idleness				
v_i	Plant	Baseline capital–electricity ratio	v_i	v_i
κ	Common	Capital-utilization elasticity	$\widehat{\kappa}$	$\widehat{\kappa}$

Capital Idleness Shifts CES Distribution Parameters

$$Y_{it} = \Omega_{it}\left(\alpha_K \widetilde{K}_{it}^{\rho} + \alpha_L L_{it}^{\rho} + \alpha_M M_{it}^{\rho} + \alpha_E E_{it}^{\rho}\right)^{\frac{1}{\rho}},$$
$$\widetilde{K}_{it} = \left[(1-\kappa)K_{it}^{\rho} + \kappa(v_i E_{it})^{\rho}\right]^{1/\rho}, \qquad v_i = \frac{K_{i,2010}}{E_{i,2010}}.$$

Solving this shifts the cost shares of capital and electricity:

$$\widetilde{s}_{Kj} = (1-\kappa)s_{Kj}, \qquad \widetilde{s}_{Ej} = s_{Ej} + \kappa s_{Kj}.$$

Electricity now matters twice: directly in production and indirectly by keeping capital running.

	Without capital idleness	With capital idleness
Capital	s_{Kj}	$(1-\kappa)s_{Kj}$
Labor	s_{Lj}	s_{Lj}
Materials	s_{Mj}	s_{Mj}
Electricity	s_{Ej}	$s_{Ej} + \kappa s_{Kj}$

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Price Elasticities

Structural Estimation

- Electricity's share of unit cost is

$$s_{E,i}^{C} \equiv \frac{\partial \ln c_i}{\partial \ln p^E}. \quad \text{where } c_i = \frac{1}{\Omega_i}\left(\alpha_K^{\sigma} p_K^{1-\sigma} + \alpha_L^{\sigma} p_L^{1-\sigma} + \alpha_M^{\sigma} p_M^{1-\sigma} + \alpha_E^{\sigma} p_E^{1-\sigma}\right)^{\frac{1}{1-\sigma}}.$$

- CES demand and conditional factor demand imply

$$\begin{aligned} \varepsilon_{Y,i} &= -\eta s_{E,i}^{C}, \\ \varepsilon_{X,i} &= (\sigma - \eta) s_{E,i}^{C}, \qquad X \in \{L, M\}, \\ \varepsilon_{E,i} &= -\sigma + (\sigma - \eta) s_{E,i}^{C}. \end{aligned}$$

Reduced-form Analysis

- A 1 ppt increase in shock intensity decreases electricity and output by 8.9% and 1.7%
- A 1 ppt increase in shock intensity increases electricity prices by 1.12%

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